

2010376103 CSE4261 Assignment-8

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1 Training and evaluating a U_Net by your favorite dataset publicly available for Segmantion task.

[Github code link](#)

For image segmentation using the U_Net model, we chose the DRIVE dataset. The DRIVE dataset is a retinal image dataset that contains 20 images for training and 20 images for testing. Table 1 shows the evaluation metrics of U_Net on the DRIVE dataset. Figure 1 shows the segmentation result of two images.

Table 1: Performance of U-Net on the DRIVE dataset

Dataset	Accuracy	Validation Loss	Mean IoU
DRIVE	0.9497	0.1541	0.7717

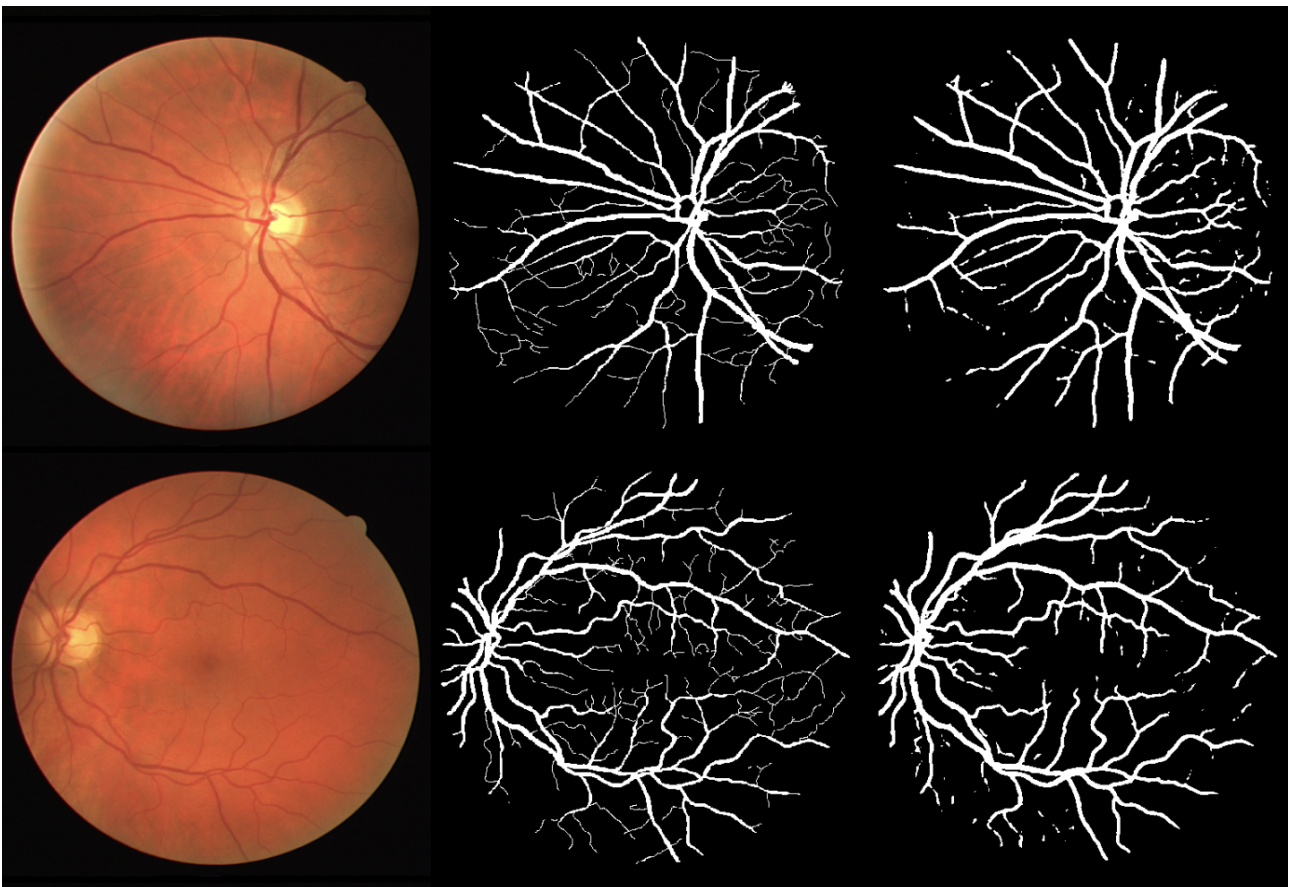


Figure 1: Retina image Segmentation

2 Training and evaluating a U-Net by your favorite dataset publicly available for Crowd Counting task

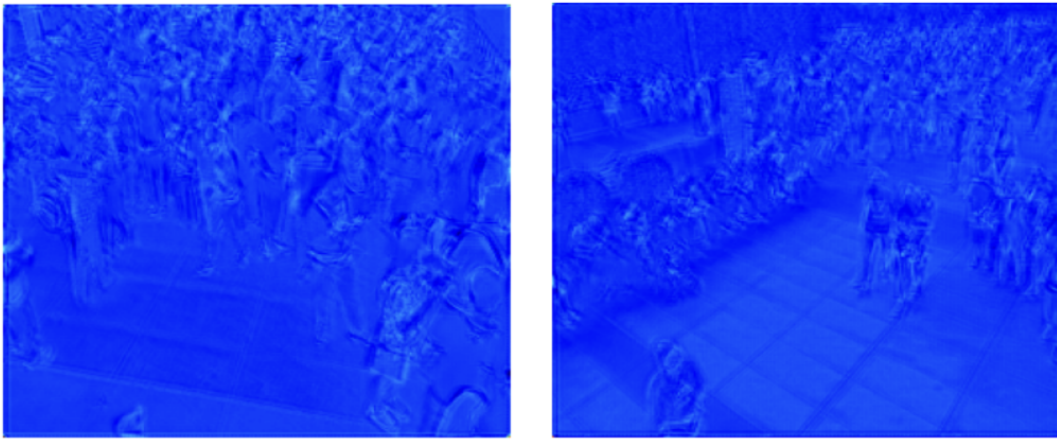
[Github code link](#)

We used the ShanghaiTech dataset to evaluate crowd counting performance using the U-Net model. For Sample 1, the predicted count was 166.82 compared to the true count of 23.00, indicating significant overestimation. In contrast, Sample 2 had a predicted count of 185.34 and a true count of 181.00, showing a much closer

match. These results highlight that while U-Net can approximate dense crowd regions well, it may struggle in sparse crowd scenarios. Figure 2 shows the prediction result of two images in prediction map.



(a) Images containing Crowd



(b) Corresponding Prediction Map of images

Figure 2: Output of Crowd Counting Model using U_Net

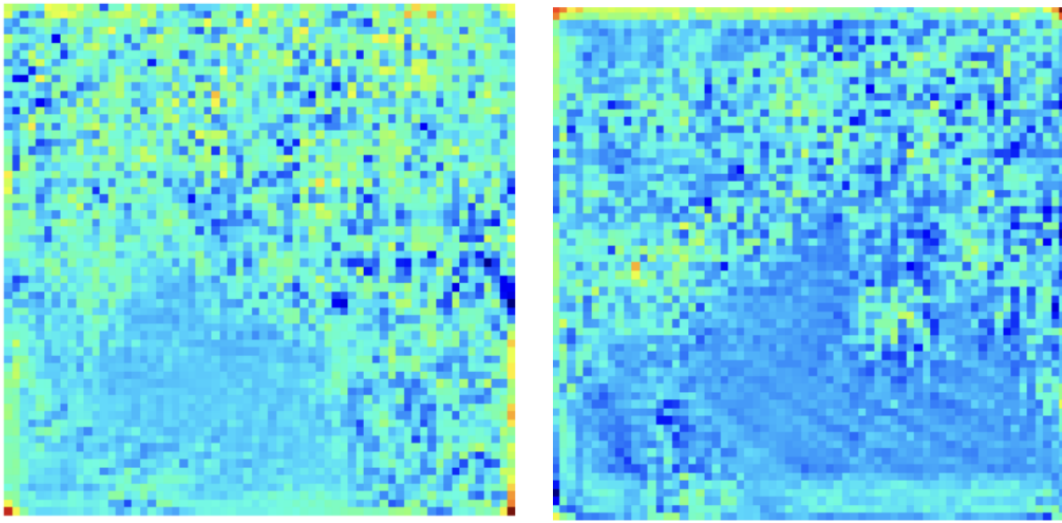
3 Training and evaluating an MCNN by your favorite dataset publicly available for Crowd Counting task

[Github code link](#)

We used the ShanghaiTech dataset to evaluate the MCNN model for crowd counting. In Sample 1, the predicted count was 6.86 compared to the true count of 23.00. Similarly, in Sample 2, the predicted count was 7.89 while the true count was 181.00. These results indicate that the MCNN model significantly underestimates crowd sizes, particularly in densely populated scenes, suggesting limitations in its ability to generalize across varying crowd densities. Figure 3 shows the prediction result of two images in prediction map.



(a) Images containing Crowd



(b) Corresponding Prediction Map of images

Figure 3: Output of Crowd Counting Model using MCNN

4 Comparing MCNN-based and U-Net based crowd counters

U-Net outperformed MCNN in crowd counting on the ShanghaiTech dataset. In dense scenes, U-Net's prediction (185.34 vs. 181.00) was much closer to the true count than MCNN's (7.89). Similarly, in sparse scenes, U-Net overestimated while MCNN consistently under-predicted. In simple terms, U-Net could guess crowd sizes more accurately than MCNN. MCNN gave much lower numbers, especially when there were many people in the image.